AYUSH THAKUR

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PROFESSIONAL SUMMARY

B.Tech Electrical Engineering student at Delhi Technological University (DTU) with a strong foundation in Java, Spring Boot, backend development, databases, and AI/ML. Experienced in developing RESTful backend services and integrating data-driven components into full-stack applications. Interested in software engineering, backend systems, and AI-driven solutions.

EDUCATION

Delhi Technological University (DTU) | (B.Tech) in Electrical Engineering | 2025 – 2029

- Academic Standing: **CGPA: 9.3 / 10.0** (8.7 First Semester | 9.9 Second Semester)
- National Rankings: Secured admission via JEE Main (AIR 29,185) and JEE Advanced (AIR 15,100).
- Relevant Coursework: Programming in Python, Engineering Mathematics, Physics, Numerical Methods.

Ramakrishna Senior Secondary School | New Delhi | Class XII (CBSE) | 2025

- Result: 96.34% in PCM (Physics, Chemistry, Mathematics).

TECHNICAL PROJECTS

DC Motor Parameter Estimation via Multi-Output Regression | Python, Scikit-Learn, Pandas

• Developed a machine learning solution to estimate six critical physical parameters of a DC motor from operational input data.
• Implemented a multi-output Random Forest Regressor with 500 estimators and a maximum depth of 15 to predict multiple physical constants simultaneously.
• Achieved an R^2 of 0.603 for Back EMF Constant estimation and an R^2 of 0.45 for Armature Resistance prediction.
• Outperformed Linear and Ridge Regression baselines through model optimization and data scaling.

FireWatch — AI-Powered Satellite Thermal Event Intelligence Platform | Java, Spring Boot, MySQL, REST APIs

• Developed the backend layer for an AI-powered satellite intelligence platform for detecting, contextualizing, classifying, and prioritizing thermal anomalies using NASA FIRMS and satellite-derived data.
• Built Spring Boot REST APIs and service-layer components for NASA FIRMS ingestion, hotspot management, analytics, GeoJSON delivery, and risk assessment.
• Implemented MySQL persistence with Spring Data JPA/Hibernate and integrated Sentinel Hub and ESA WorldCover data for environmental and land-cover feature enrichment.
• Integrated the Java backend with a Python FastAPI ML microservice for multi-class thermal anomaly classification and incorporated model outputs into operational risk scoring.

TECHNICAL SKILLS

- Languages: **Java** (Proficient), **Python** (Proficient), **C/C++** (Intermediate).
- Backend Development: **Spring Boot, Spring Security, REST APIs, JDBC, Hibernate, JPA, Servlets, JSP, Maven, Gradle, Spring AI**.

Libraries & ML: **NumPy, Pandas, Matplotlib, Scikit-Learn; Supervised Learning, Regression, Multi-Output Models, Feature Engineering, Model Evaluation (RMSE, R^2).**

- Databases & Tools: **MySQL, MongoDB, Git, JUnit, Log4j, Docker, Cloud Deployment, Microservices, Aws, Linux, Ansible, Jenkins, Terraform.**

ACHIEVEMENTS & HONORS

- JEE Main 2025: Secured All India Rank 29,185 (Top 2% out of ~1.4 million candidates).
- JEE Advanced 2025: Secured All India Rank 15,100.
- Academic Merit: Graduated Class XII with 96.34% in PCM (94.8% overall).

INTERESTS

- AI in Power Electronics & Renewable Energy
- Competitive Programming